

A RESPONSE THEORY FOR THE RIEMANN ZERO GAS: AMPLITUDE CONSERVATION, A THERMAL BRAGG MIRROR, AND THE TWO-SUM INTERFERENCE

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ABSTRACT. Two companion notes measured how the local statistics of Riemann zeros respond to the prime oscillations of the explicit formula: an amplitude channel $w(\tau)$ and a zero-displacement channel $v(\tau)$, functions of $\tau = \log p/L$, $L = \log \frac{t}{2\pi}$. Here we organize those measurements into a coherent response theory — after first identifying, mechanically explaining, and correcting a *sampling-grid systematic*: the gap midpoints at which all wave phases are evaluated are themselves displaced by the prime waves, so regressions on restricted frequency bases transfer omitted-line content into the measured coefficients. A ground-truth test on the real grid shows a four-prime basis biases w downward by 0.14–0.24 in absolute units (16–40% relative), while a complete basis over all prime powers recovers the truth to $\leq 1.5\%$; every number below uses the complete-basis convention. (i) The two channels obey a tight linear constraint, $\sqrt{w} = 1.017 - 0.884v$ (residual RMS 0.0075; slope ± 0.021 under window-block bootstrap), which extends to prime powers: the gap-conditioned contents fall on the same τ -curve as the primes. The constraint survives an out-of-sample test on unseen primes; among equal-parameter alternatives intensity-unitarity fails by a factor 6, while the $\sqrt{w}$ -linear and w -linear forms are at present statistically indistinguishable. (ii) Extending the measurements to $\tau = 0.61$ shows that w *changes sign* at $\tau_0 = 0.447 \pm 0.005$, with response phases that jump from 0 to π without rotating (quadrature below 8% through the crossing): both channels are real, consistent with a lossless (reactive) medium. (iii) A thermal Bragg model *reproduces* the crossing: the ideal mirror sits at the Riemann–Siegel fold $\tau = 1/2$; dressing it with the measured zero-displacement jitter and the measured beyond-fold plateau places the zero at 0.443, within 1σ of the measurement, with unit coherent amplitude ($A = 1.02$). (iv) The structure factor $(1 - 2\tau)$ is derived in three lines as the pair count of the folded main sum; pointwise $|Z|^2$ regressions — dense-grid measurements immune to the sampling systematic — instead follow the shifted-second-moment family $\approx 2(1 - \tau)$ through and beyond the fold, so that the complementary *mirror channel* is a *tent* peaked at the fold: the approximate functional equation made audible as an interference channel. (v) Stripping the measured waves leaves a core correlation $r^* = 0.977$, insensitive to the basis; the finer decomposition of the residual variance and the standing-wave refinement of the thermal model are quoted under the previous convention and flagged for re-measurement.

1. INTRODUCTION

The first companion note [5] measured the joint law of consecutive-zero gaps and between-zero maxima of Hardy’s Z -function and its deviation from CUE characteristic polynomials. The second [6] dissected that deviation into explicit-formula channels: each prime wave $\cos(t \log p)$ enters the amplitudes at full weight and splits between a direct amplitude channel w and a zero-displacement channel v , both collapsing onto one-variable laws in the Berry variable $\tau = \log p/L$ [2]; the channel decomposition itself is an observable-level counterpart of the hybrid Euler–Hadamard product [3]. This note asks what *physics* organizes those curves, and answers with five measured structures: a two-channel linear constraint, a sign-crossing,

a thermal Bragg model that reproduces the crossing’s location, an arithmetic derivation of the structure factor, and the measurement of the complementary mirror channel. A sixth structure imposed itself during the final audit: the measurement grid itself responds to the primes, and Section 2 documents the systematic and the corrected convention now used throughout. Throughout, the style of the companions is kept: no closed forms are fitted; every ingredient is either measured or derived; and every claim is accompanied by the test that tried to kill it.

Data and conventions are those of [5, 6]: eleven amplitude windows spanning $t = 1.2 \times 10^5$ to 2.7×10^{11} (the last from Odlyzko’s table of zeros near the 10^{12} -th [4]), plus six low windows down to $t \approx 1.2 \times 10^3$ for the extended- τ measurements; channels are measured by linear regression of $\log(\text{unfolded maxima})$ and $\log(\text{unfolded gaps})$ on $\cos(t_n \log p)$, $\sin(t_n \log p)$ at gap midpoints, with own-gap controls for w (the unfolded gap $\tilde{\delta}_n = g_n L / 2\pi$ and $\tilde{\delta}_n^2$ as covariates). All channel coefficients are measured with *complete* regression bases (Section 2).

2. THE SAMPLING-GRID SYSTEMATIC

All per-gap channel measurements in this program regress log-maxima and log-gaps on wave columns evaluated at gap midpoints. The midpoints are determined by the zeros alone — exogenous to the amplitudes — but they are not a passive grid: the zeros are displaced by the prime waves (the v channel itself), so the sampling pattern carries the primes. On the real midpoints a wave column at one frequency acquires a small, systematic correlation with the columns of *other* frequencies (Gram $R^2 \approx 7 \times 10^{-4}$ against the small-prime block for a near-fold column; identically zero on the smooth Riemann–von Mangoldt grid). A regression on a restricted basis therefore transfers omitted-line content into the retained coefficients, and the per-prime normalization $w = \text{coefficient} \times p^{1/2}$ amplifies the transfer by $p^{1/2}$, so the bias grows toward the fold.

Two measurements make this quantitative. A basis ladder on the near-fold band shows the transferred share converging as the basis completes: the apparent beyond-fold plateau -0.354 under a band-only basis shrinks to -0.042 under the full sub-band basis and is stable under further extension. And a ground-truth test — synthetic amplitudes with known line content built on the *real* midpoint grid — shows a four-prime basis under-measures w by 0.14 – 0.24 in absolute units (16–40% relative; 20 – 22σ), while the complete basis recovers the truth to $\leq 1.5\%$.

Convention adopted throughout this note and the revised companion: all channel coefficients are measured with complete bases — every prime power p^k with $\log p^k / L \leq 0.45$ (to 0.55 for the extended- τ curve), capped at $q \leq 720$. The $360 \rightarrow 720$ cap step moves the constraint slope below by $\sim 1\%$, which we quote as the residual basis-cutoff systematic. Three classes of results are immune by construction: dense-grid pointwise regressions (the tent of Section 7), regression-free statistics (the dephasing scale s_{eff}), and residual correlations (the stripped core r^* , which moves only from 0.9783 to 0.9774). The first companion note contains no channel regressions and is unaffected. One limitation is recorded: the ground-truth test uses a single window, a single noise draw, and i.i.d. noise with a pure line spectrum; a broadband grid-correlated component — hinted at by non-null chirp-sum controls in the project log — could in principle bias even complete bases, and its characterization is open.

3. THE TWO-CHANNEL CONSTRAINT

The second companion's sum rule — total prime content of unit weight — suggested that the two channels share a budget. Under the complete-basis convention the sharing is a strikingly tight linear constraint.

Observation 1 (Two-channel constraint). *Across 44 matched (w, v) pairs ($4 \text{ primes} \times 11 \text{ windows}$, t from 10^5 to 2.7×10^{11}), the free fit gives $\sqrt{w} = 1.017 - 0.884v$ with residual RMS 0.0075 (slope ± 0.021 under window-block bootstrap), at the per-prime scatter floor. Constraining the intercept to 1 worsens the fit (RMS 0.0102): the intercept exceeds unity by $\approx 1.7\%$ — several times the cap-step effect on the intercept ($+0.3\%$) — and its status, physical or residual systematic, is left open.*

The constraint extends beyond the primes. Measuring the prime-power lines under the same (gap-conditioned, complete-basis) convention and normalizing by the explicit-formula weight $p^{-k/2}/k$ gives 0.700 ± 0.002 , 0.560 ± 0.003 , 0.530 ± 0.002 , 0.338 ± 0.004 , 0.335 ± 0.006 for $q = 4, 8, 9, 25, 27$ — each matching the prime curve $w(\tau)$ at that line's own $\tau = \log p^k/L$ at the 1–2% level (a small systematic excess, $2\text{--}4\sigma$ at $q = 4$, is left open). The gap-conditioned transmission law is a law of prime powers, governed by τ alone; the *unconditioned* contents, by contrast, sit at unity for all primes (revised companion, sum rule).

Stress tests (Figure 1). *Out of sample*: w for $p = 11, 13$ — not used in the fit, extending the v -range — follows the free line with RMS 0.0269 over 22 pairs, with a one-sided mean offset of -0.018 : the constraint survives, though less crisply than under the previous convention; we suspect genuine curvature at larger τ . *Form race*: among equal-parameter alternatives the intensity-unitarity form $w + cv^2 = 1$ fails by a factor 6 in RMS; the $\sqrt{w}$ -linear and w -linear forms, by contrast, are at present statistically indistinguishable (0.0127 vs 0.0099 in the w -scale), so the data no longer single out amplitude over intensity as the linearly shared quantity — an honest retreat from the previous version of this note.

A cautionary record. Under the previous (restricted-basis) convention the fit gave $\sqrt{w} = 0.990 - 1.025v$, the constrained slope 1.068 matched a quadrature phase-bookkeeping ratio $\beta/2 = 1.07$, and we presented that agreement as a motivated coefficient. The corrected slope 0.884 ± 0.021 does not match it. The bookkeeping coincidence was an artifact of the sampling-grid bias — precisely the kind of trap this project's audit culture exists to catch, and we record it as such.

4. THE SIGN CROSSING AND LOSSLESSNESS

Extending the channel measurements to the primes up to 31 (eleven primes) over twelve windows (132 points, τ up to 0.61, all complete-basis) shows that the amplitude channel does not merely shrink — it *crosses zero* at

$$\tau_0 = 0.447 \pm 0.005$$

(statistical, from a window-block bootstrap of the linear fit over the $\tau \in [0.30, 0.58]$ points; the linear-fit-zero definition carries a fit-range systematic of ± 0.009 , which a quadratic-fit zero removes — stable at 0.447–0.448 across ranges; the crossing position is convention-sensitive at the 0.05 level: in the restricted-basis convention of the earlier version it appeared near 0.40, and an independent re-implementation with a single $\log \tilde{\delta}$ control likewise finds ≈ 0.40 — the value 0.447 refers to the complete basis with the $(\tilde{\delta}, \tilde{\delta}^2)$ controls) and settles on a shallow negative plateau: beyond the crossing, peaks are slightly *low* where the prime wave

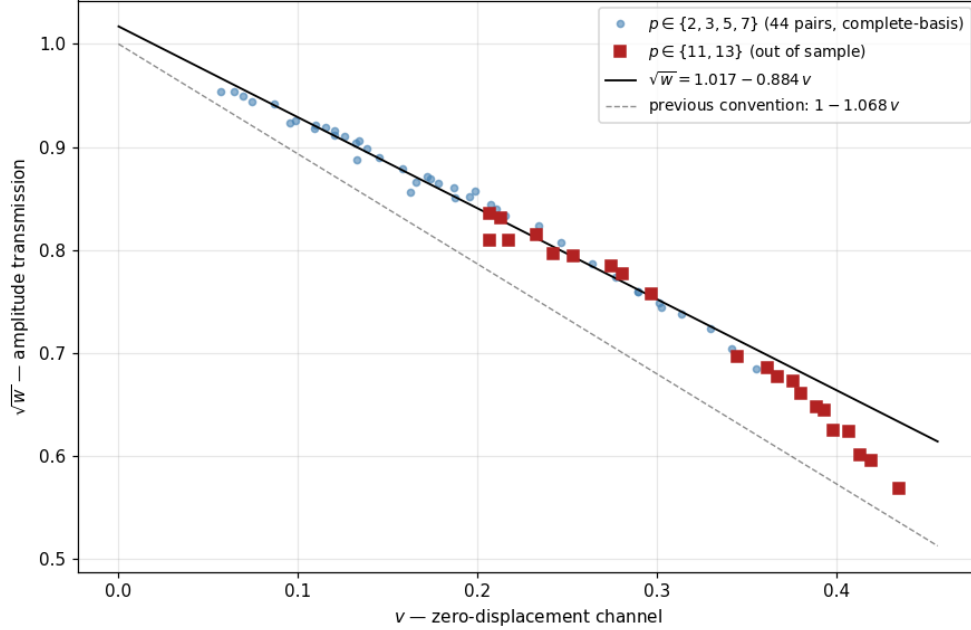


FIGURE 1. The two-channel constraint under the complete-basis convention. Blue: the 44 pairs of the fit. Red: the 22 out-of-sample pairs ($p = 11, 13$). Solid: the corrected line $1.017 - 0.884 v$; dashed: the previous convention’s line.

is high. The plateau depth is -0.04 to -0.09 depending on how far the covariate basis is extended beyond the fold — much shallower than the -0.16 of the previous convention, most of which the audit exposed as basis transfer. Two qualifications, sharpened by the audit: the statement “peaks are low where the wave is high” holds *at fixed observed gap* — without the own-gap controls the unconditional near-fold response is positive and of order unity — and the crossing lies at 1.6σ from the coarse exhaustion-analysis crossover $\tau^* = 0.40 \pm 0.03$ of [6]; the identification of τ^* with the zero of w remains plausible but awaits a complete-basis re-measurement of the exhaustion curve.

The response phases are diagnostic. Through the crossing the phase of w jumps from ≈ 0 to $\approx \pi$ with quadrature-to-in-phase ratios at or below 0.011 for $\tau < 0.35$, 0.08 through the crossing region, and 0.14 beyond; the phase of v stays pinned at 0 . A damped resonance would rotate through $\pm\pi/2$; a real response that changes sign is the signature of a *lossless* (conservative, Hermitian) medium. The curtain does not absorb the prime waves; it redistributes and reflects them — consistent, at the level of response theory, with a Hermitian generator underneath.

5. THE THERMAL BRAGG MIRROR

Where should the crossing be? The Riemann–Siegel main sum contains only $n \leq N = \sqrt{t/2\pi}$, i.e. $\log n \leq L/2$: the amplitude has a sharp spectral fold at $\tau = 1/2$, and a wave whose period matches two lattice spacings ($1/\tau = 2$) is precisely the Bragg condition of the zero lattice. The ideal mirror sits at $\tau = 1/2$; the measured crossing sits at 0.447 . The shift is thermal.

Model: $w(\tau) = A(1 - 2\tau)e^{-c\tau^2} [\tau < 1/2] + B$, where $(1 - 2\tau)$ is the ideal structure factor (derived in Section 6), B is the indirect channel measured as the beyond-fold plateau

($B = -0.092$ on the corrected curve), and the Gaussian factor is Debye–Waller dephasing by the zeros’ displacement jitter. The jitter is *measured*: comparing gap midpoints with the Riemann–von Mangoldt flow gives σ_u per window; in unfolded units it grows from 0.198 to 0.272 across our windows — the crystal warms logarithmically, as the classical $S(t)$ variance suggests — giving $c = (2\pi\sigma_u)^2/2 = 0.77\text{--}1.46$, mean 1.17.

Observation 2 (The crossing is reproduced). *With c fixed from the measured jitter and B fixed from the measured plateau, the single remaining parameter fits to $A = 1.02$ (unit coherent amplitude), and the model’s zero lands at $\tau = 0.443$ — within 1σ of the measured 0.447 ± 0.005 (Figure 2). Two qualifications: across the plateau’s convention range (-0.04 to -0.09) and the jitter’s window spread the model zero moves over $0.44\text{--}0.47$, comfortably containing the measurement; and since B is measured from the same curve’s beyond-fold points, the agreement is partly internal consistency rather than prediction. Under the previous convention the model and the measurement disagreed by more than the thermal family’s internal spread, and the previous version of this note could claim only a bracket $[0.37, 0.41]$; the sampling-grid correction moved the measured crossing onto the model. Nothing about the model changed — only the measurement.*

Two refinements of the previous version are flagged rather than asserted: the window-temperature collapse of the fixed- τ scatter (to ± 0.003) and the damping-scale race that preferred the standing-wave factor were both computed on the previous convention’s curve, and await re-measurement on the corrected one. The model-independent ingredient stands: measured as the ratio of log characteristic functions of the displacement series, with no Gaussian assumption and no regression,

$$s_{\text{eff}} = \frac{\ln |\langle e^{2iku} \rangle|}{\ln |\langle e^{iku} \rangle|} = 4.11 \pm 0.04,$$

the Gaussian value 4 dressed by the measured sub-Gaussian displacement statistics (excess kurtosis -0.3 to -0.5 — level repulsion suppressing the tails).

6. THE STRUCTURE FACTOR, DERIVED

Derivation 3. Write $Z = 2 \sum_{n \leq N} n^{-1/2} \cos \phi_n$, $\phi_n = \theta(t) - t \log n$. In $|Z|^2$ the frequency $\log p$ is produced exactly by the pairs (m, mp) with $mp \leq N$, each contributing weight $4(m \cdot mp)^{-1/2}$, while the DC part is $2 \sum_{n \leq N} n^{-1}$. Hence the relative modulation is

$$R_{\text{direct}}(p) = 2p^{-1/2} \frac{S_1(N/p)}{S_1(N)}, \quad S_1(x) = \sum_{m \leq x} \frac{1}{m},$$

which is $p^{-1/2} \cdot 2(1 - 2\tau)$ in the continuum limit and vanishes identically for $p > N$.

The structure factor $(1 - 2\tau)$ of the Bragg model is therefore not a guess: it is the pair-counting arithmetic of the folded sum, with a sharp horizon at the fold.

7. THE MIRROR CHANNEL: A TENT AT THE FOLD

Testing the derivation pointwise — regressing $|Z(t)|^2$ on the prime waves over fine grids (1.3×10^6 points per window, two windows) — produced a surprise: the measured modulation

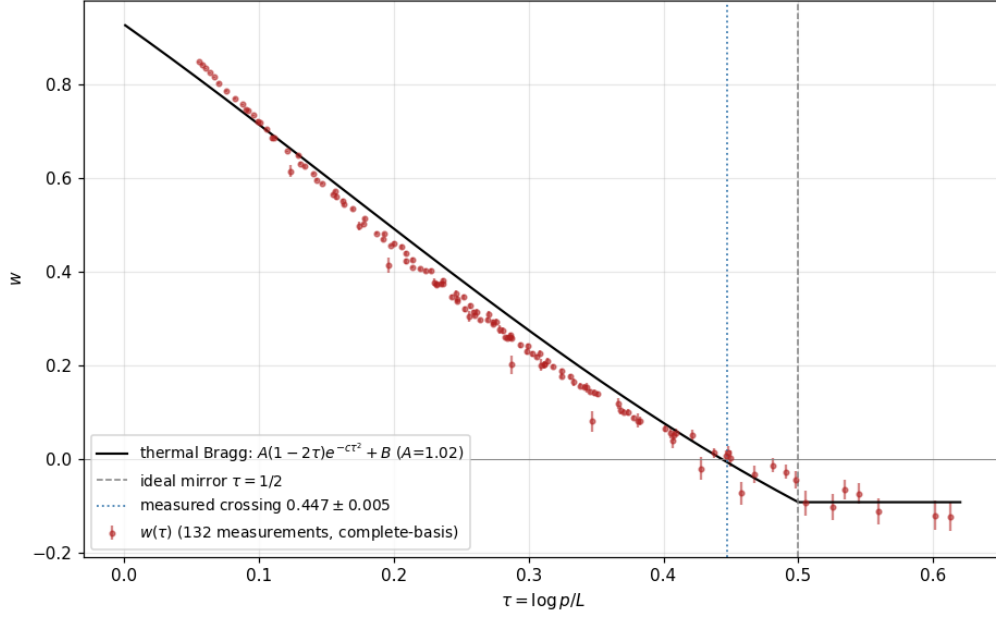


FIGURE 2. The sign crossing on the complete-basis curve, with the thermal Bragg model: c from the measured jitter, B from the measured plateau, $A = 1.02$ the only fitted number. The model's zero (0.443) lands within 1σ of the measured crossing (0.447 ± 0.005).

does *not* vanish at the fold. It follows, through and beyond the fold, the shifted-second-moment family

$$R(p) \approx 2p^{-1/2} \frac{L - \log p + c_0}{L + c_0} \approx p^{-1/2} \cdot 2(1 - \tau),$$

dying only at the full horizon $p \approx t/2\pi$ (cf. the classical mean-value theorems for $\int |\zeta|^2(m/n)^{it}$ [1]). The difference

$$R_{\text{mirror}} = R_{\text{measured}} - R_{\text{direct}}$$

is a *tent*: it rises as 2τ below the fold, peaks at the fold with value 1, and descends as $2(1 - \tau)$ beyond (Figure 3). Its mechanism is the chirp cross-terms $\cos(2\theta(t) - t \log nm)$ — precisely the interference between the two mirror-image sums of the approximate functional equation — which lock onto frequency $\log p$ at the stationary-phase products $nm \approx (t/2\pi)/p$. Beyond the fold, where no direct pairs exist, everything one hears is the mirrored sum: the functional equation of ζ , measured as an interference channel. The parameter-free tent sits systematically below the two-window measurements by 6–20%; the exact normalization awaits the true main-term constants (the free fit prefers $c_0 \approx 1.1$ over the naive $2\gamma - 1$). Measurements at $\tau \gtrsim 0.62$ are excluded: the zero-spacing comb band sets in there, and non-prime control frequencies reach the tent scale — unlike the max-channel measurements, these pointwise regressions do not yet carry a full placebo battery, which we flag as a limitation. Being dense-grid pointwise measurements, they are immune to the sampling-grid systematic of Section 2.

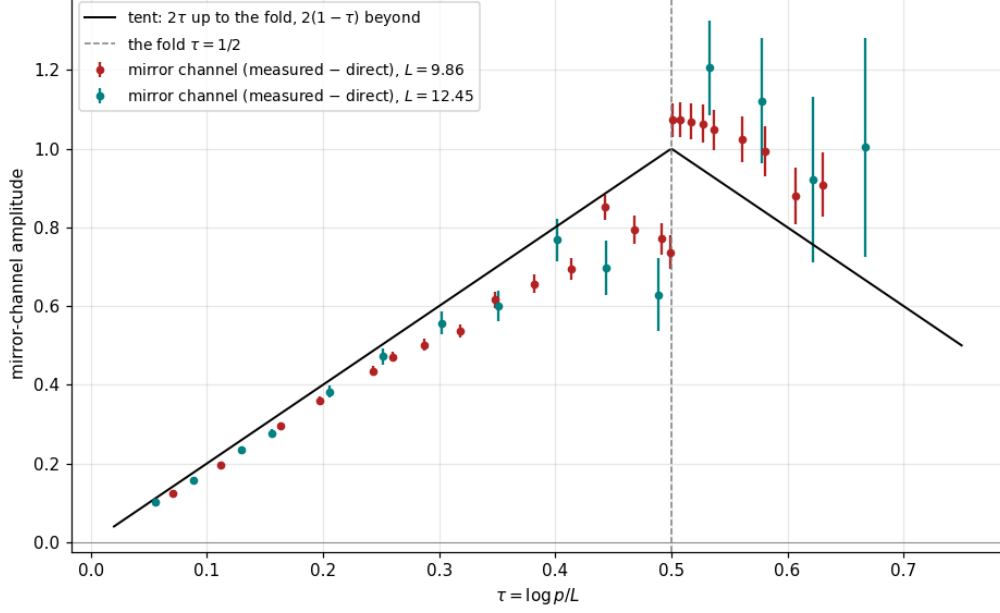


FIGURE 3. The mirror channel at two windows against the parameter-free tent. The direct channel dies at the fold; the mirror channel peaks there.

8. THE RESIDUAL VARIANCE, DECOMPOSED

The companion note's residual amplitude variance $V_{\text{res}}(L) = 0.163 \rightarrow 0.244$ and its decomposition — $\sim 4.5\%$ unstripped tail, a gap-modulated interaction share of 25–45%, and a core growing at 0.035 per unit L , the CUE rate at one tenth the amplitude — were measured under the previous convention. The headline that survives the audit unchanged is the correlation statistic: stripping the measured waves from both series leaves a core pair correlation $r^* = 0.9783$ under the previous basis and 0.9774 under the complete basis — *insensitive* to the convention, as a correlation of residuals should be. The variance rates and the interaction share are flagged for re-measurement; we retain the qualitative statement that a small intrinsic component grows at close to the random-matrix tempo.

9. SYNTHESIS AND OPEN PROBLEMS

The measured response theory of the zero gas now reads: each prime-power wave enters at full explicit-formula weight and splits between transmission and zero displacement along a tight linear constraint ($\sqrt{w} = 1.017 - 0.884v$; the $\sqrt{w}$ -versus- w form question is open); the medium is consistent with lossless, with a Bragg mirror at the Riemann–Siegel fold whose observed position 0.447 is reproduced by dressing the ideal mirror with the measured thermal jitter; the ideal structure factor is the fold's pair-counting arithmetic; beyond the fold the response is carried by the functional equation's mirror sum, whose interference channel is a tent peaked at the fold; the stripped core correlation 0.977 is basis-insensitive; and the measurement grid itself is part of the medium — the zeros' displacement field couples every restricted regression to the omitted lines.

Open problems: (1) the exact tent normalization from the classical shifted-moment main terms, including the χ -side; (2) the mapping of the pointwise mirror channel to the corrected max-channel plateau (-0.04 to -0.09): sign and size; (3) partial-reflection dressing $r(\tau)$

below the Bragg condition (a first 2θ -locked candidate at the few-percent level is recorded in the project log); (4) a derivation of the constraint's slope 0.884 and intercept 1.017; (5) the distribution of the intrinsic core — is it GUE?; (6) universality: does the same response theory hold for Dirichlet L -functions?; (7) a mechanistic theory of the sampling-grid coupling — the grid the zeros provide is not a passive ruler.

REPRODUCIBILITY

All measurements are scripts 62–84 in <https://github.com/azizran/riemann-zero-statistics> (directory `qm_riemann/`; archived at Zenodo, doi:10.5281/zenodo.22942475), runnable from the stored datasets; the dated research log records the failed intermediate models alongside the results.

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Computational assistance, literature search and drafting: Claude (Anthropic). An independent re-implementation of the core measurements, written from scratch with a different AI assistant (DeepSeek) and run on the raw Odlyzko tables (repository folder `bagimsiz_dogrulama/`) reproduced the direct $|Z|^2$ channel ($r = 0.9989$ over 21 primes below the fold), the sign change of w without rotation (quadrature $\leq 5\%$), and — once the $q^{-1/2}$ normalization is matched — the $2(1 - \tau)$ tent family to within 1–8%. Errors are the author's.

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